EMTH210
LECTURE 32:

Announcement:
Recaps last lecture
Refer to table of Laplace Transform Rules

6.1 LAPLACE TRANSFORM

- important when dealing with things in electrical

The most important use of the *Laplace Transform* is in solving differential equations. The Laplace Transform, or LT, turns the problem of solving an ODE into one of solving an algebraic equation, then looking things up in a table.

purpose of it

The Laplace Transform turns a formula in terms of t into a formula in terms of s, and the inverse Laplace Transform does the opposite: , so transform, rearrange, then transform back

s is something else (not really physically intuitive)

fancy L: $\mathcal{L}$

transform of a function, to give us another function

$$F(s) = \mathcal{L}\{f(t)\} \qquad \text{and} \qquad f(t) = \mathcal{L}^{-1}\{F(s)\}$$

a transformation not a function

t is physically meaningful (usually time)

We will see the definition and theory of the Laplace Transform soon, but first we will look at how it is used in practice.

One powerful new capability Laplace Transforms give us is the ability to solve differential equations involving piecewise functions, such as terms that get switched on or off at particular times. This happens frequently in real-world applications when power to a system or part of a system is switched on or off, or when a valve is opened or closed, or when a physical connection is made or broken.

$y(t) \to Y(s)$

The basic idea behind our use of LTs to solve ODEs is that we transform the ODE (which has an unknown function of t and its derivatives) into an algebraic equation (in terms of an unknown function of s), rearrange it a bit, and then use the *inverse* Laplace transform to turn the algebraic equation (in s) into an algebraic equation (in t) — the solution to the ODE. Moreover, the method automatically inserts initial conditions.

via table

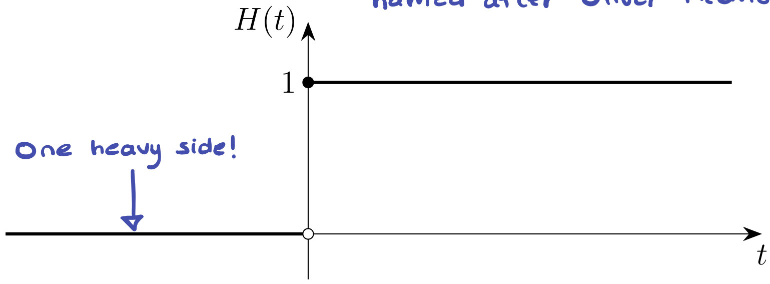
$Y(s) \to y(t)$

6.2 HEAVISIDE STEP FUNCTION

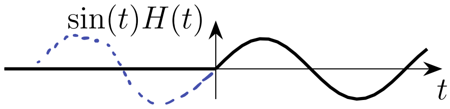
To work with piecewise functions using Laplace Transforms, we need to write them as single formulas using a function called the *Heaviside step function*[1]:

named after Oliver Heaviside

$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$



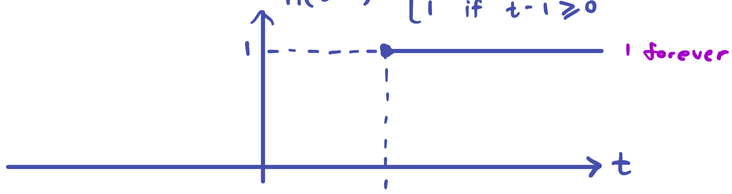
This function acts like a switch: anything we multiply by it gets multiplied by either 0 or 1, depending on where it is evaluated. E.g. $\sin(t)H(t)$ looks like this:



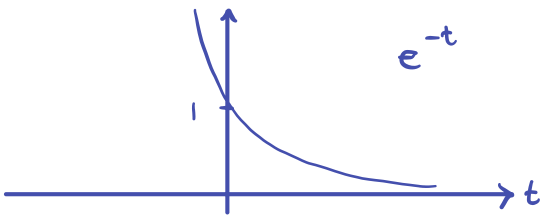
Example 6.2.1

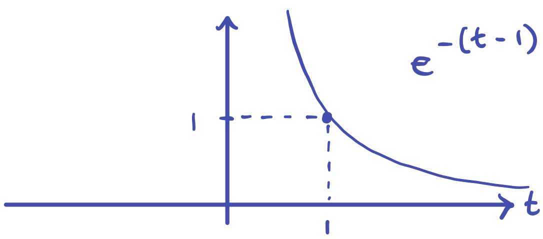
Sketch the graph of $f(t) = e^{-(t-1)}H(t-1)$.

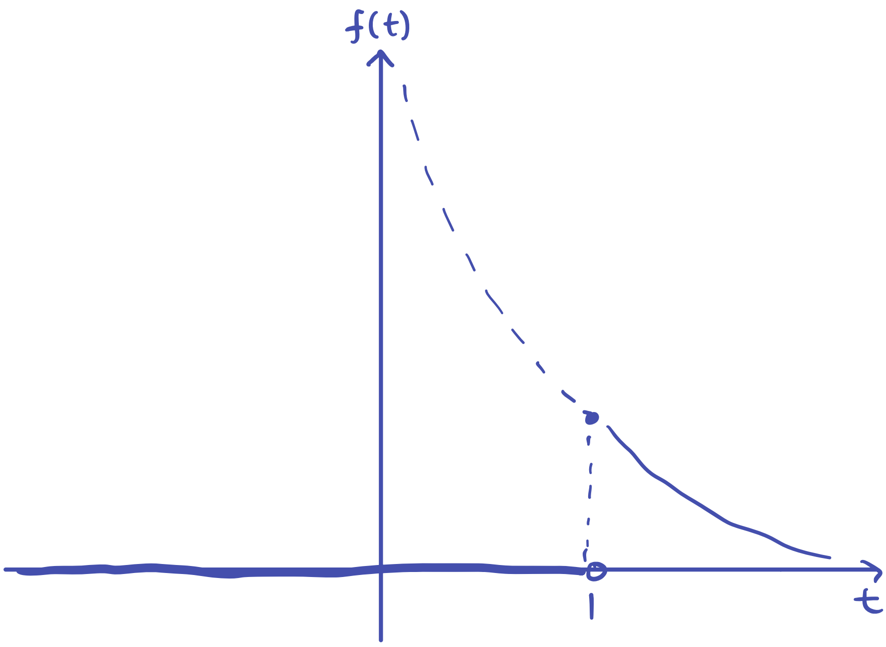
$$H(t-1) = \begin{cases} 0 & \text{if } t-1 < 0 \\ 1 & \text{if } t-1 \geq 0 \end{cases}$$



Solution.







♦

[1]Sometimes called the *unit* step function, $u(t)$ or $U(t)$.

L32 starts here

6.3 SOLVING AN ODE USING LAPLACE TRANSFORMS

$y(t) \to Y(s)$

1. LT the ODE using linearity, inserting the initial conditions. $y(0),\ y'(0)$

$Y(s) \to y(t)$

2. Rearrange the resulting algebraic equation, solving for the transformed function (usually $Y(s)$).
3. Take the inverse LT of the resulting algebraic equation, by looking things up in a table and using linearity.

Example 6.3.1

Use the table to solve the following first order ODE using Laplace transforms.

$$y' - y = e^{-(t-1)}H(t-1) \qquad \text{where} \qquad y(0) = 2.$$

Solution.

1. $\mathcal{L}\{y'-y\} = \mathcal{L}\{e^{-(t-1)}H(t-1)\}$

Laplace Transform of both sides equal

$y = y(t)$

Let $Y(s) = \mathcal{L}\{y(t)\}$ (always)

often not written down

CONVENTION WE WILL FOLLOW

Formula Sheet:

Linearity $\Rightarrow \mathcal{L}\{y'\} - \mathcal{L}\{y\} = \mathcal{L}\{\underbrace{e^{-(t-1)}}_{g(t-a)} H(t-1)\}$, $a = 1$

$\mathcal{L}\{y'\} = sY(s) - y(0)$

$Y(s)$ Defined

Second shift theorem:

$\mathcal{L}\{g(t-a)H(t-a)\} = e^{-as}G(s)$

instead of putting a shift in a function, we are shifting out

$G(s) = \mathcal{L}\{g(t)\}$

$\Rightarrow$ need $g(t)$

$g(t-1) = e^{-(t-1)}$

unshift: $g(t) = e^{-t} = e^{(-1)t}$

table: $G(s) = \frac{1}{s-(-1)} = \frac{1}{s+1}$

$\Rightarrow \mathcal{L}\{e^{-(t-1)}H(t-1)\} = e^{-s}\frac{1}{s+1}$

usually the longest step

2. $sY(s) - 2 - Y(s) = e^{-s}\frac{1}{s+1}$

$(s-1)Y(s) = e^{-s}\frac{1}{s+1} + 2$

want to keep rearranging until we find something in our table

$Y(s) = e^{-s}\frac{1}{(s+1)(s-1)} + \frac{2}{s-1}$

$Y(s) = e^{-s}\frac{1}{s^2-1} + 2\frac{1}{s-1}$

usually would need partial fractions decomposition

we can take inverse LP of both sides

$\mathcal{L}^{-1}\{e^{-as}G(s)\} = g(t-a)H(t-a)$

$G(s) = \frac{1}{s^2-1} \Rightarrow g(t) = \sinh(t)$

$g(t-1) = \sinh(t-1)$

♦

3. $\therefore y(t) = \sinh(t-1)H(t-1) + 2e^t \quad t \geq 0$

6.4 LAPLACE TRANSFORM DEFINITION AND THEORY

The definition and for the Laplace Transform of a function $f(t)$ when $t \geq 0$ is:

$$F(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t = \mathcal{L}\{f(t)\}.$$

Example 6.4.1

Use the definition of the Laplace Transform to find the Laplace Transform of $f(t) = 1$.

Solution.

$F(s) = \mathcal{L}\{f(t)\}$ ← always true! Standard convention: $\mathcal{L}\{g(t)\} = G(s)$

$= \mathcal{L}\{1\}$ $\qquad \mathcal{L}\{y(t)\} = Y(s)$

$\vdots$

$= \int_0^\infty (1)e^{-st}\,dt$ ← integrating wrt. t, so s is constant

$= \left[\frac{-1}{s}e^{-st}\right]_{t=0}^{\infty}$

$= \lim_{\ell\to\infty}\left[\frac{-1}{s}e^{-st}\right]_{t=0}^{\ell}$

$= \frac{-1}{s}\left(\lim_{\ell\to\infty} e^{-s\ell} - e^0\right)$

if $s > 0$

$= \frac{-1}{s}(0 - 1)$

$= \frac{1}{s}$

♦

NB: If $s \leqslant 0$ the integral is infinite and so this $F(s)$ is only defined for $s > 0$.

A function $f(t)$ has a Laplace transform provided:

1. $f(t)$ is piecewise continuous on $t \geq 0$; ← necessary in this whole topic
2. $f(t)$ is exponentially bounded for $t \geq 0$. Formally, this means that there exists M, c such that

$$|f(t)| < Me^{ct}, \qquad \forall t \geq 0.$$

Under these conditions the Laplace Transform exists for $s > c$. An example of a function that fails the second condition is e^{t^2}.

Since the Laplace Transform is only used for problems where $t \geq 0$ (i.e. time starts at 0), another use of the Heaviside step function is to deal with functions $f(t)$ which are defined for $t < 0$ as well as for $t \geq 0$. Because the definition of the LT only considers $f(t)$ in $t \geq 0$, and ignores anything that happens in $t < 0$, for a function which has other undesired behaviour in $t < 0$, we use the Heaviside step function to "switch it off" for negative t. It is always safest to consider the LT of $\underbrace{f(t)H(t)}_{\text{switches on } f(t) \text{ at } t=0}$ rather than just $f(t)$. For example

$$\mathcal{L}\{1 \cdot H(t)\} = \int_0^\infty 1 \cdot H(t)\, e^{-st}\, \mathrm{d}t = \frac{1}{s},$$

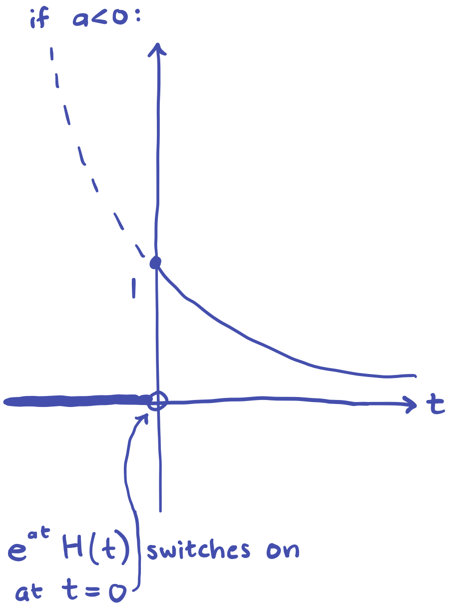
H(t) = 1 within integral range

as before.

Similarly,

$f(t) = e^{at}$

$$F(s) = \mathcal{L}\{e^{at}H(t)\} = \int_0^\infty e^{at}\, e^{-st} H(t)\,\mathrm{d}t \qquad H(t) = 1 \text{ here since } t \geqslant 0 \text{ in this integral}$$



$$= \lim_{\ell\to\infty} \int_0^\ell e^{(a-s)t}\,dt$$

$$= \lim_{\ell\to\infty} \left[\frac{1}{a-s} e^{(a-s)t}\right]_{t=0}^{\ell}$$

$$= \frac{1}{a-s}\left(\lim_{\ell\to\infty} \cancel{e^{(a-s)\ell}}^{\,0} - \cancel{e^{0}}^{\,1}\right) \qquad \text{provided } s > a,$$

We need $a-s<0$ for this to converge

$$= \frac{1}{s-a}.$$

The fact that s must be greater than a in this example is crucial, and is related to the second of the two conditions on page 6.